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DSA0601 Data Handling and Visualization

Slot A

CO2 – Assessment Test 1 (AT1)

Chart Construction Test

Instructions:

For each question, write the program in the specified language (R or Python), generate the required visualization, and answer the interpretation sub-questions clearly with justification.

CO2_AT1 – Chart Construction Test

Integrated Disciplines

Each question combines **Data Visualization + R Programming + another discipline** to promote interdisciplinary learning.

Question 1 (Healthcare Analytics)

Dataset

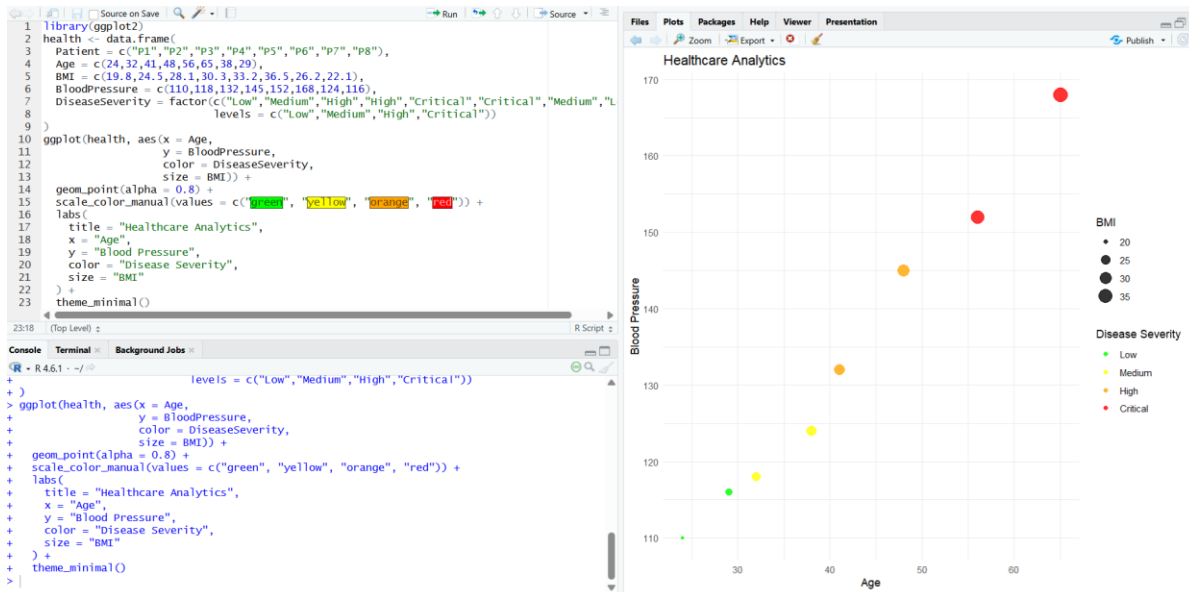
Patient	Age	BMI	Blood Pressure	Disease Severity
P1	24	19.8	110	Low
P2	32	24.5	118	Medium
P3	41	28.1	132	High
P4	48	30.3	145	High
P5	56	33.2	152	Critical
P6	65	36.5	168	Critical
P7	38	26.2	124	Medium
P8	29	22.1	116	Low

Task

Construct a visualization in **R (ggplot2)** that:

- Maps **Age** to the x-axis.
- Maps **Blood Pressure** to the y-axis.
- Represents **Disease Severity** using an appropriate color palette.
- Represents **BMI** using point size.
- Justify why your aesthetic mappings and scales effectively communicate health risk.

Interdisciplinary Integration: Healthcare + Medical Statistics + Data Visualization



Justification

- Age is mapped to the x-axis because it is an independent numerical variable.
- Blood Pressure is mapped to the y-axis to observe how it changes with age.
- Disease Severity is represented by color (Green → Low, Yellow → Medium, Orange → High, Red → Critical), making risk levels easy to distinguish.
- BMI is represented by point size, so patients with higher BMI appear as larger points.
- A scatter plot is appropriate because it effectively shows the relationship between two continuous variables while displaying additional information through color and size.

Question 2 (Climate Science)

Dataset

Month	Solar Radiation	Temperature (°C)	Rainfall (mm)
Jan	180	20	25
Feb	200	23	18
Mar	240	29	12
Apr	280	35	8
May	320	39	5

Jun	300	36	65
Jul	260	32	150
Aug	250	31	170
Sep	240	30	120
Oct	220	28	85
Nov	190	24	45
Dec	170	21	30

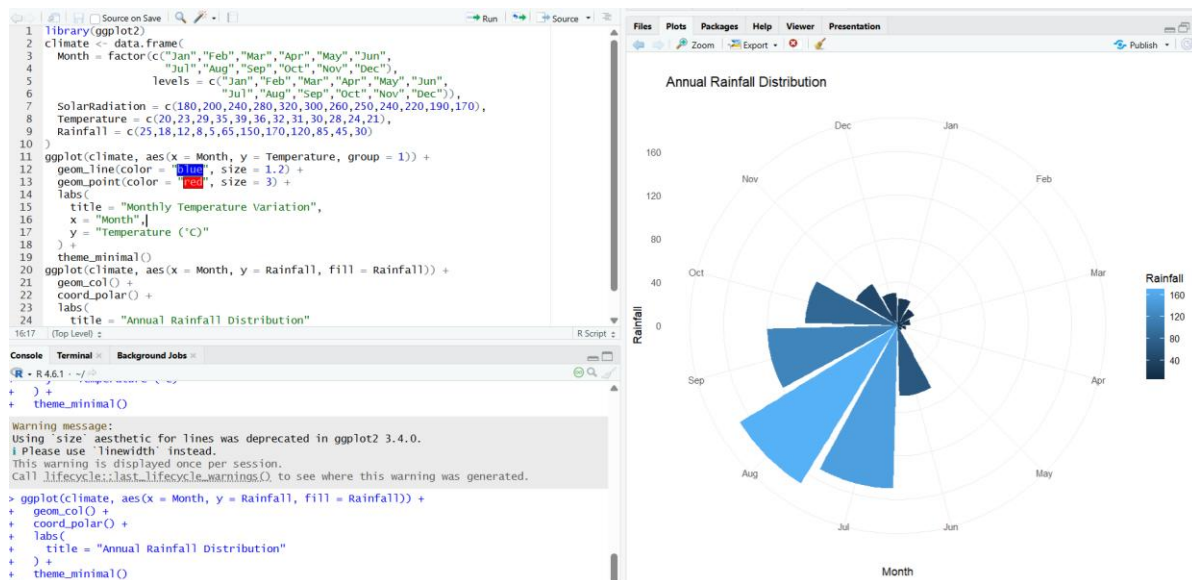
Task

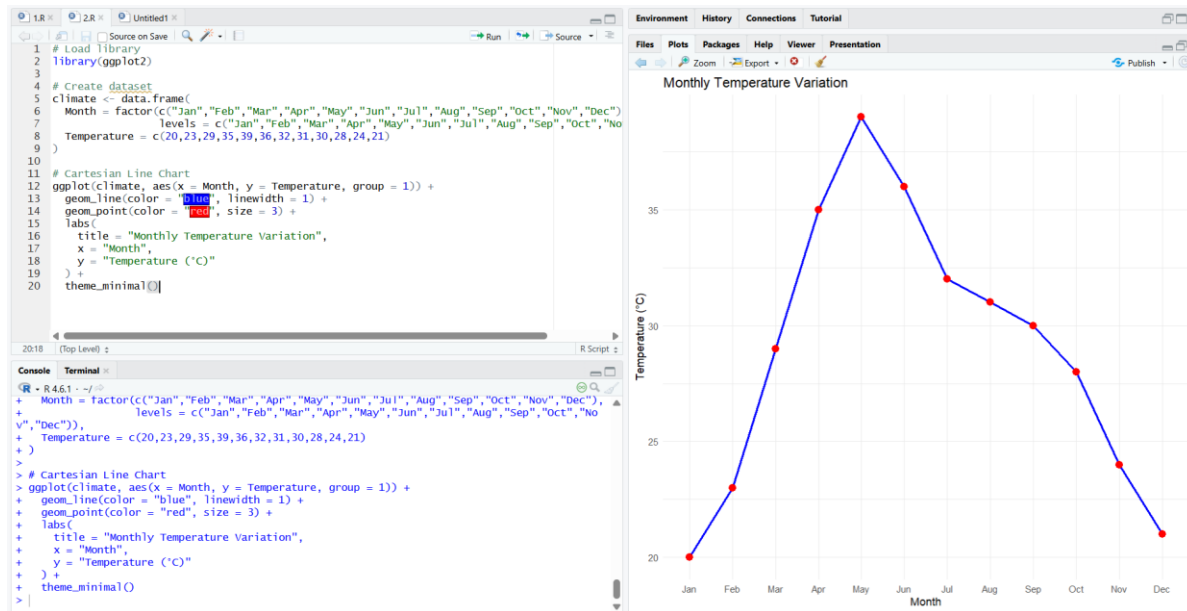
Develop **two visualizations**:

1. A Cartesian line chart showing temperature variation.
2. A Polar coordinate chart representing annual rainfall.

Compare the advantages and limitations of Cartesian and Polar coordinate systems for climate visualization.

Interdisciplinary Integration: Environmental Science + Mathematics + Data Visualization





Cartesian Coordinate System	Polar Coordinate System
Uses X-axis and Y-axis.	Uses angle and radius.
Best for showing trends over time.	Best for showing cyclic or seasonal patterns.
Easy to compare exact values.	Easy to visualize yearly or circular data.
Suitable for temperature variation.	Suitable for annual rainfall distribution.
Less attractive for seasonal data.	Exact value comparison is more difficult.

Advantages

Cartesian Line Chart

- Clearly shows temperature changes month by month.
- Easy to identify increasing and decreasing trends.
- Accurate comparison of values.

Polar Coordinate Chart

- Displays rainfall in a circular (seasonal) pattern.
- Highlights yearly cycles effectively.
- Attractive and intuitive for climate data.

Limitations

Cartesian Chart

- Does not emphasize seasonal or cyclic patterns.

Polar Chart

- Difficult to compare exact rainfall values.
- Less suitable for detailed numerical analysis.

Question 3 (Business Analytics & Marketing)

Dataset

Product	Sales (₹ Lakhs)	Profit (%)	Market Share (%)
A	210	18	24
B	185	14	20
C	260	25	28
D	145	12	15
E	120	10	13

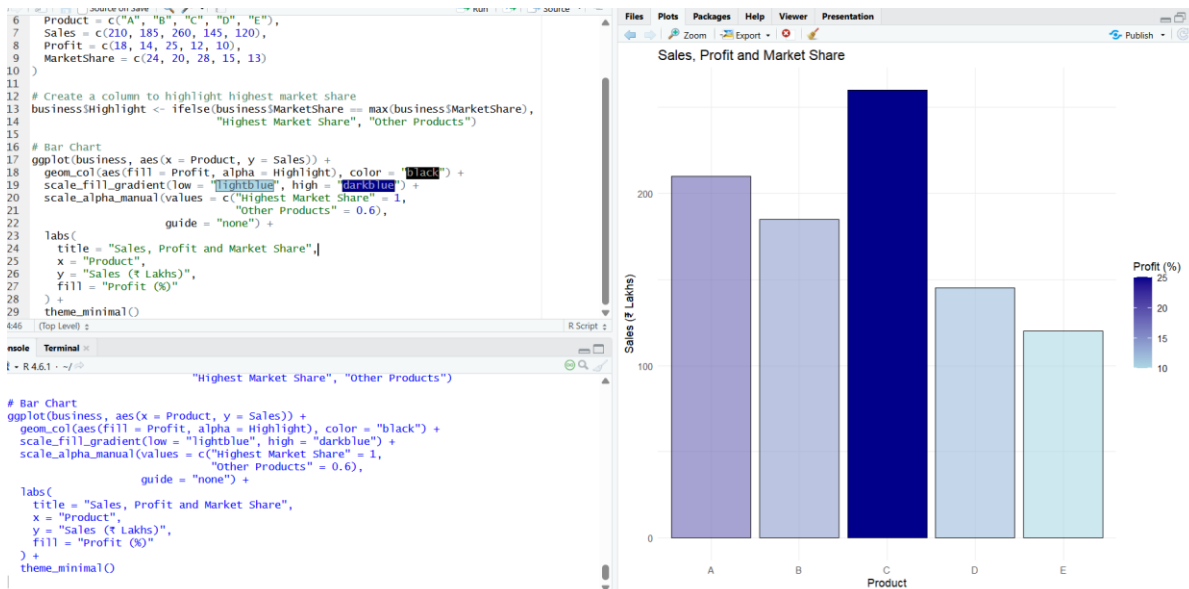
Task

Construct a visualization where:

- Sales are represented by bar height.
- Profit percentage is represented using a sequential color scale.
- Highlight the product with the highest market share using a contrasting color.

Explain why your chosen color palette improves business decision-making.

Interdisciplinary Integration: Marketing + Business Intelligence + Visualization



Justification

- Bar height represents Sales, making it easy to compare product performance.
- A sequential color scale (light blue to dark blue) represents Profit (%). Darker bars indicate higher profit.
- Product C is highlighted because it has the highest market share (28%), making it

stand out from the other products.

- The color palette helps managers quickly identify high-sales, high-profit, and market-leading products, supporting better business decisions.

Question 4 (Transportation Engineering)

Dataset

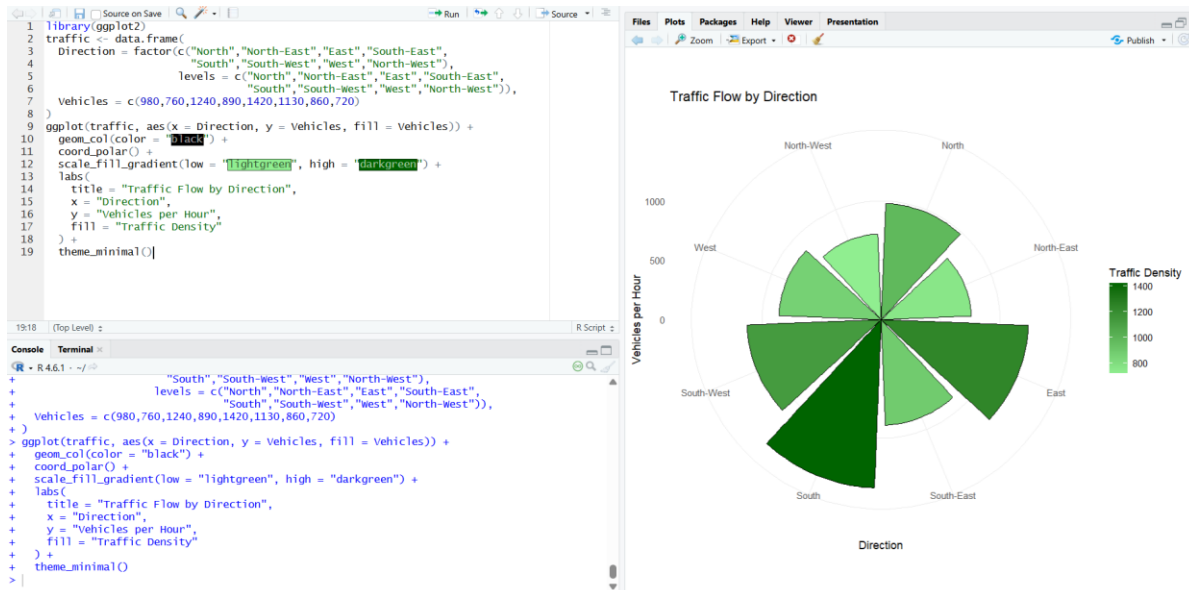
Direction	Vehicles/hour
North	980
North-East	760
East	1240
South-East	890
South	1420
South-West	1130
West	860
Direction	Vehicles/hour
North-West	720

Task

Using **R Programming**:

- Construct a **Polar Coordinate Chart** representing traffic flow.
- Apply an appropriate color palette to distinguish traffic density.
- Explain why a Polar chart is more suitable than a Cartesian bar chart for directional traffic analysis.

Interdisciplinary Integration: Civil Engineering + GIS + Transportation Analytics



Justification

Polar Coordinate Chart is used because the data represents traffic flow in different directions (360°).

- The bar length represents the number of vehicles per hour.
- A green sequential color palette is used to represent traffic density:
 - Light Green → Low traffic
 - Dark Green → High traffic

Question 5 (Environmental Sustainability Dashboard)

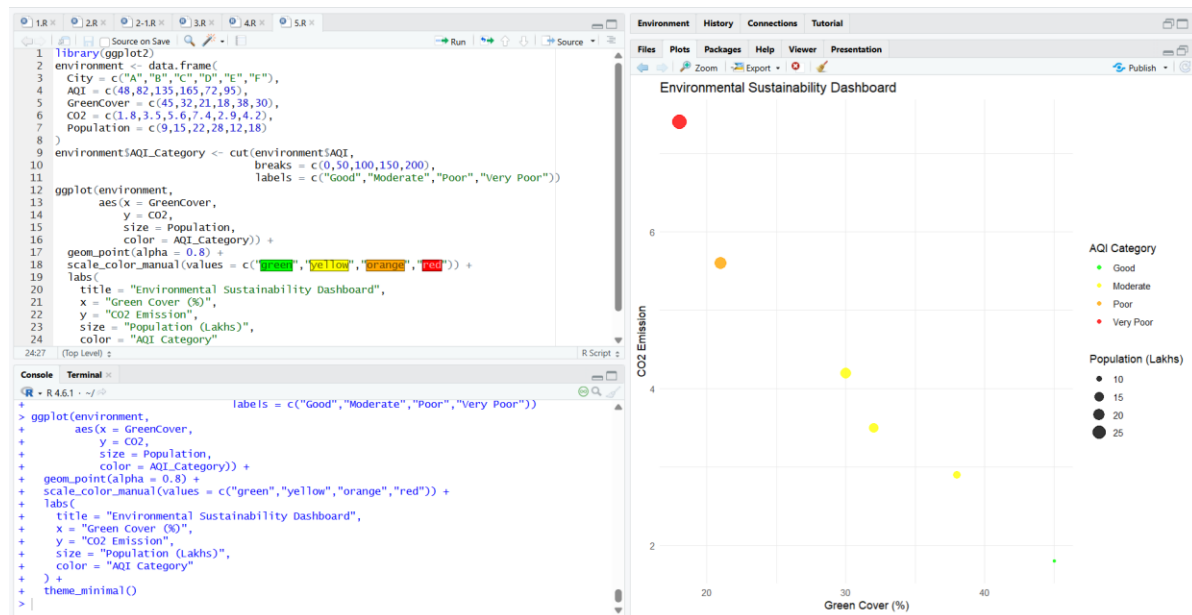
Dataset

City	AQI	Green Cover (%)	CO ₂ Emission	Population (Lakhs)
A	48	45	1.8	9
B	82	32	3.5	15
C	135	21	5.6	22
D	165	18	7.4	28
E	72	38	2.9	12
F	95	30	4.2	18

Task

Design a visualization dashboard that includes:

- Appropriate aesthetic mappings.
- Suitable scales for continuous variables.
- An effective color palette for AQI categories.
- Justification for every visualization design choice.



Justification

Aesthetic Mappings

- X-axis → Green Cover (%)
- Y-axis → CO₂ Emission
- Point Size → Population (Lakhs)
- Point Color → AQI Category

Suitable Scales

- Continuous scales are used for Green Cover, CO₂ Emission, and Population.
- A categorical color scale is used for AQI Categories.